# Hard Problems in Lattice-Based Cryptography - Introduction to Lattice-based Cryptography

## Worse-case to Average-case Reduction

what's the worst case of SVP?

## Shortest Vector Problems(SVP)

In the first post, we talked about the first successive minimum $\lambda_1$ and good and bad bases. This problem is based on the hardness of finding $\lambda_1$ given a bad basis.

Given an arbitrary basis $B$ of some lattice $L = L(B)$, find a shortest nonzero lattice vector, i.e a $\mathbf{v} \in L$ for which $||v|| = \lambda_1(L)$.

### Approximate Short Vector Problem($SVP_\gamma$)

Given a basis $\mathbf{B}$ of an n-dimensional lattice $L = L(\mathbf{B})$, find a nonzero vector $\mathbf{v} \in L$ for which $||v|| \leq \gamma(n).\, \lambda_1(L)$.

### Decisional Approximate SVP($GapSVP_\gamma$)

### Closest Vector Problems (CVP)

### Bounded Distance Decoding Problem

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## Modern Hard Problems

### Short Integer Solution(SIS)

### Inhomogeneous SIS

SIS and ISIS can be reduced to GapSVP

### Learning With Errors (LWE)

#### Search LWE

Decision LWE

Ring LWE